Lecture 16

Pollution and health

Dana Andersen
AEM 6510

Roadmap

- How do we estimate the effect of a policy or environmental change when the treated and control groups do not (counterfactually) look the same in the cross-section?
- What is the mortality cost of lead?

Difference-in-differences

Our comparisons so far

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1. Comparing two groups with random assignment to treatment **(RCT)**
2. Comparing two groups where there is a local discontinuity (i.e. discrete change) in policy **(regression discontinuity)**

In both of these we are comparing groups in the **cross-section**: there is no concept of time, before and after a policy was enacted, etc

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This assumption is often hard to defend[1]

Let's try to relax this assumption by exploiting **temporal comparisons** in addition to the cross-sectional comparison

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Difference-in-differences

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The estimated effect of a policy is simply the difference in expected outcomes between treated and control groups:

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$$\delta = E[Y^1|D=1] - E[Y^0|D=0]$$

It's exactly that for an RCT since D was randomly assigned, and its the difference in conditional expectations (conditional on being around the threshold) for RDD

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The name comes from the fact that we are taking the difference (3) between two differences (1 and 2)

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The identifying assumption required is less strict than for difference approaches

DD assumption: the treatment and control group would have followed **parallel trends** in the absence of treatment

- i.e. the difference in outcomes would have remained constant

This is much less stringent than requiring the outcomes to have been the same in the absence of treatment

Difference-in-differences

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Example: Suppose we want to understand the effect of a conservation policy passed in New York on biodiversity

Suppose also that:

- The effect of the New York policy is given by **B**
- Each state has it's own fixed determinants of biodiversity (e.g. land cover, average temperature, etc) given by **NY, PA, MA, etc**
- Each period has it's own determinants of biodiversity, common across all states (e.g. federal policy, global climate change) given by $\mathbf{T_0}$, $\mathbf{T_1}$, where 0 is years before the policy is passed, and 1 is after

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1. Compare New York to another state after the policy is passed
2. Compare New York to itself, before and after the policy is passed

The cross-sectional difference

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If we take a simple difference across states, we can't disentangle whether the difference is due to the policy **B** or differences in these other factors **NY - PA**

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There are other determinants of biodiversity that are different before and after the policy that are **not** the New York policy: federal policy changes, trends in urbanization and pollution

If we take a simple difference over time, we can't disentangle whether the difference is due to the policy **B** or differences in other factors that are changing over time $T_1 - T_0$

Difference-in-differences

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We take the time series differences for NY and PA:

	After	Before	Time Series Difference
New York	$B + NY + T_1$	$NY + T_0$	$(B + NY + T_1) - (NY + T_0) = \mathbf{B + T_1 - T_0}$
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Next, difference the time series differences to get the DD[1]

[1] You can also difference in the opposite order and end up with the same result

Difference-in-differences

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The time series differences lets us control for all fixed determinants within a state (**NY**)

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The cross-sectional difference lets us control for all period-specific determinants common across all states (**T_1**)

Combining these two differences addresses both and lets us recover **B**, the true effect of the policy!

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2. Time-varying, but common across all states

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It only can address determinants of biodiversity that are either:

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2. Time-varying, but common across all states

If there is a determinant of biodiversity that is varying over time, and differentially across states, DD will fail to correctly estimate **B**

- State climate trends, state pollution trends, etc

Difference-in-differences

Suppose that there is another determinant of biodiversity C^{NY}_1, C^{NY}_0 that only occurs in New York and varies over time

- e.g. climate in New York relative to Pennsylvania

Our DD is then

	After	Before	Time Series Difference
New York	$B + NY + T_1 + C^{NY}_1$	$NY + T_0 + C^{NY}_0$	$(B + NY + T_1) - (B + NY + T_1) = \mathbf{B + T_1 - T_0 + C^{NY}_1 - C^{NY}_0}$
Pennsylvania	$PA + T_1$	$PA + T_0$	$(PA + T_1) - (PA + T_0) = \mathbf{T_1 - T_0}$
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Pennsylvania	$PA + T_1$	$PA + T_0$	$(PA + T_1) - (PA + T_0) = \mathbf{T_1 - T_0}$
		Difference-in-differences:	$\mathbf{(B + T_1 - T_0) - (T_1 - T_0) = B + C^{NY}{}_1 - C^{NY}{}_0}$

DD cannot isolate the effect of **B** versus $\mathbf{C^{NY}{}_1 - C^{NY}{}_0}$

There cannot be any (uncontrolled for) time varying differences between NY and PA if we want to correctly estimate **B**

Difference-in-differences: to the data

Now lets see how this works in practice: notebook here

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What we are going to do is create a fake dataset where we **know the true value of what we want to estimate**

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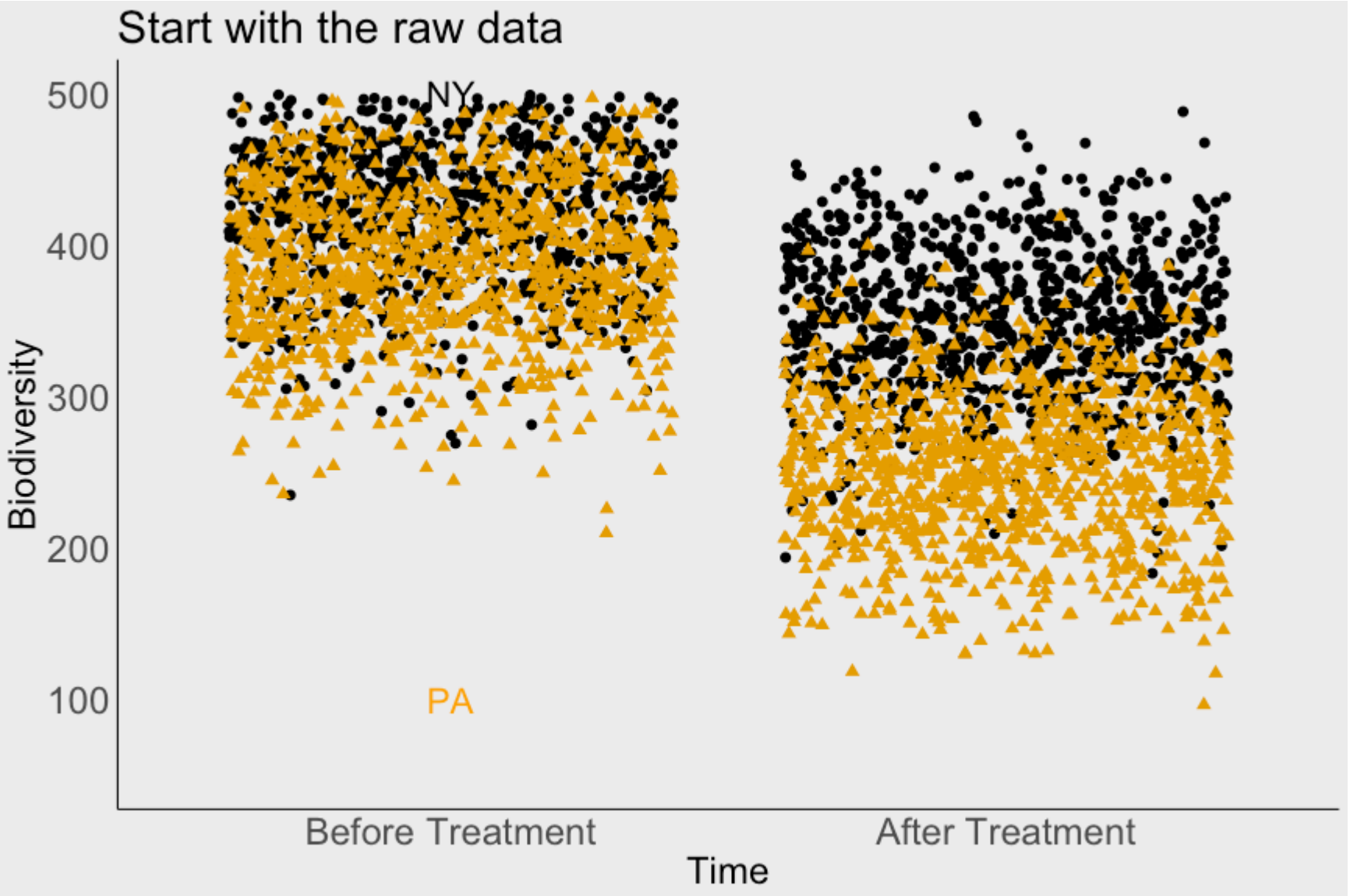
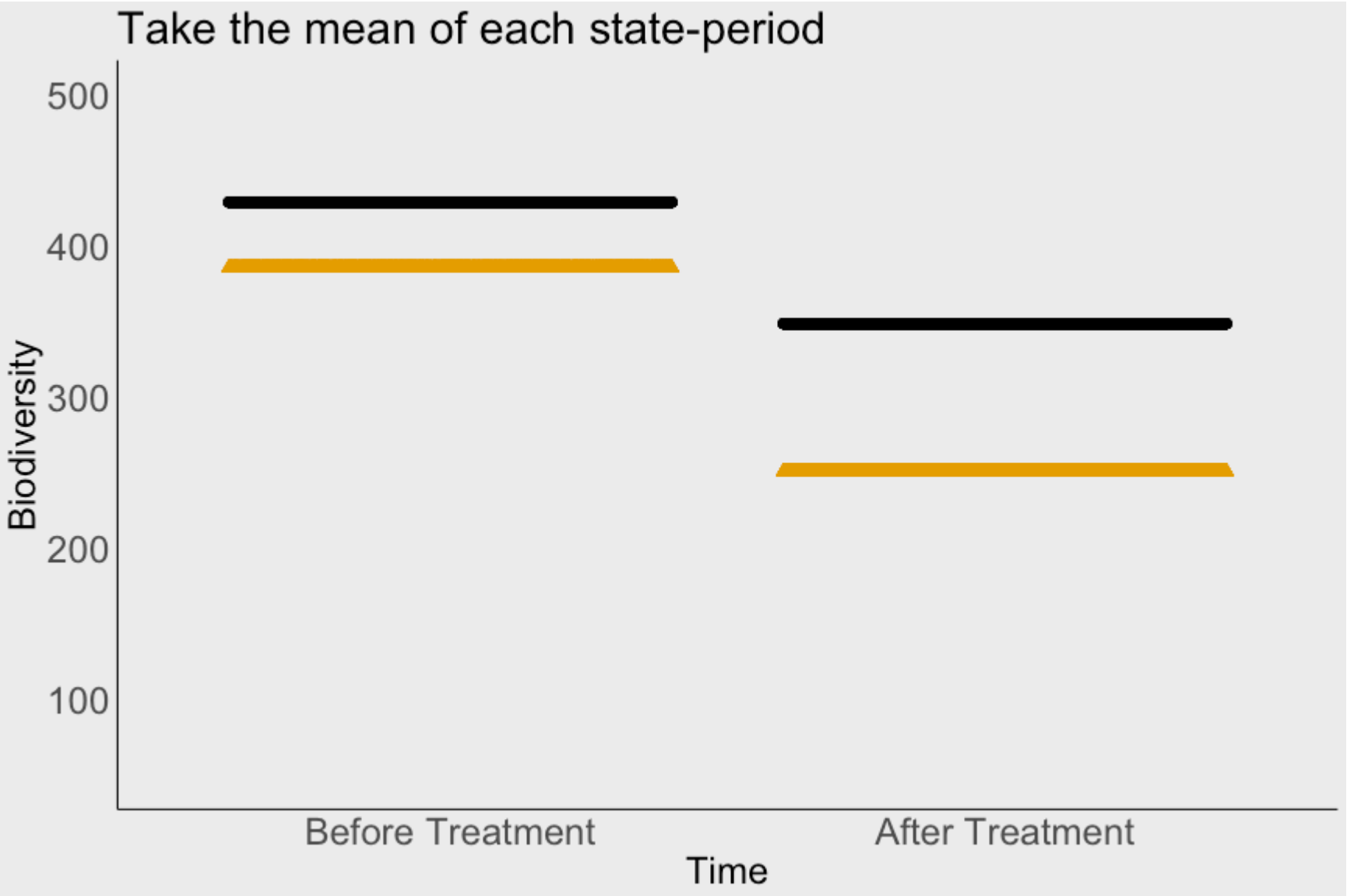
This is a good skill to practice to make sure you understand methods and that your code works correctly

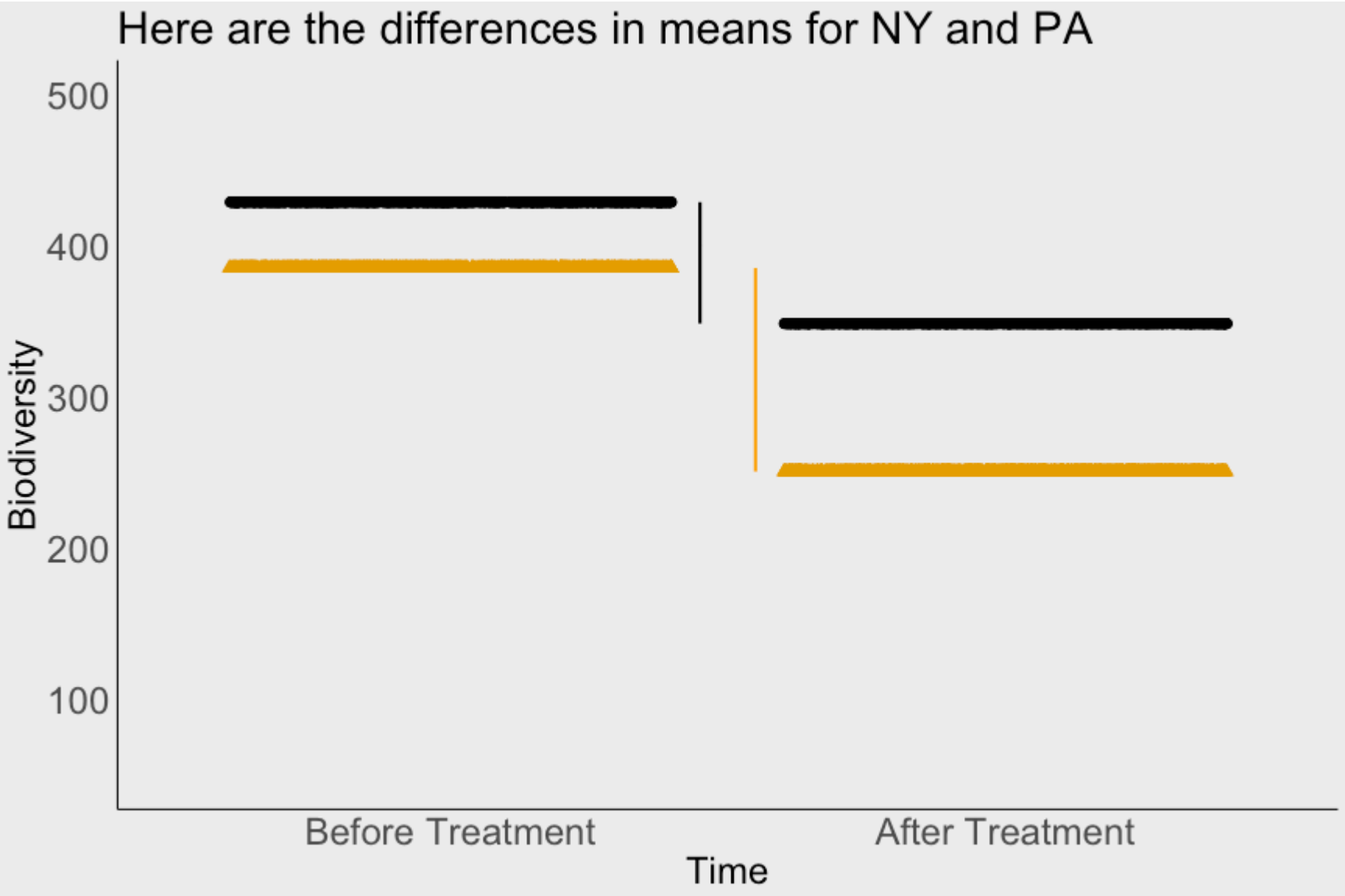
DD step 1


Start with the raw data
500
400
300
200
100
Biodiversity
NY
PA
Before Treatment
After Treatment
Time

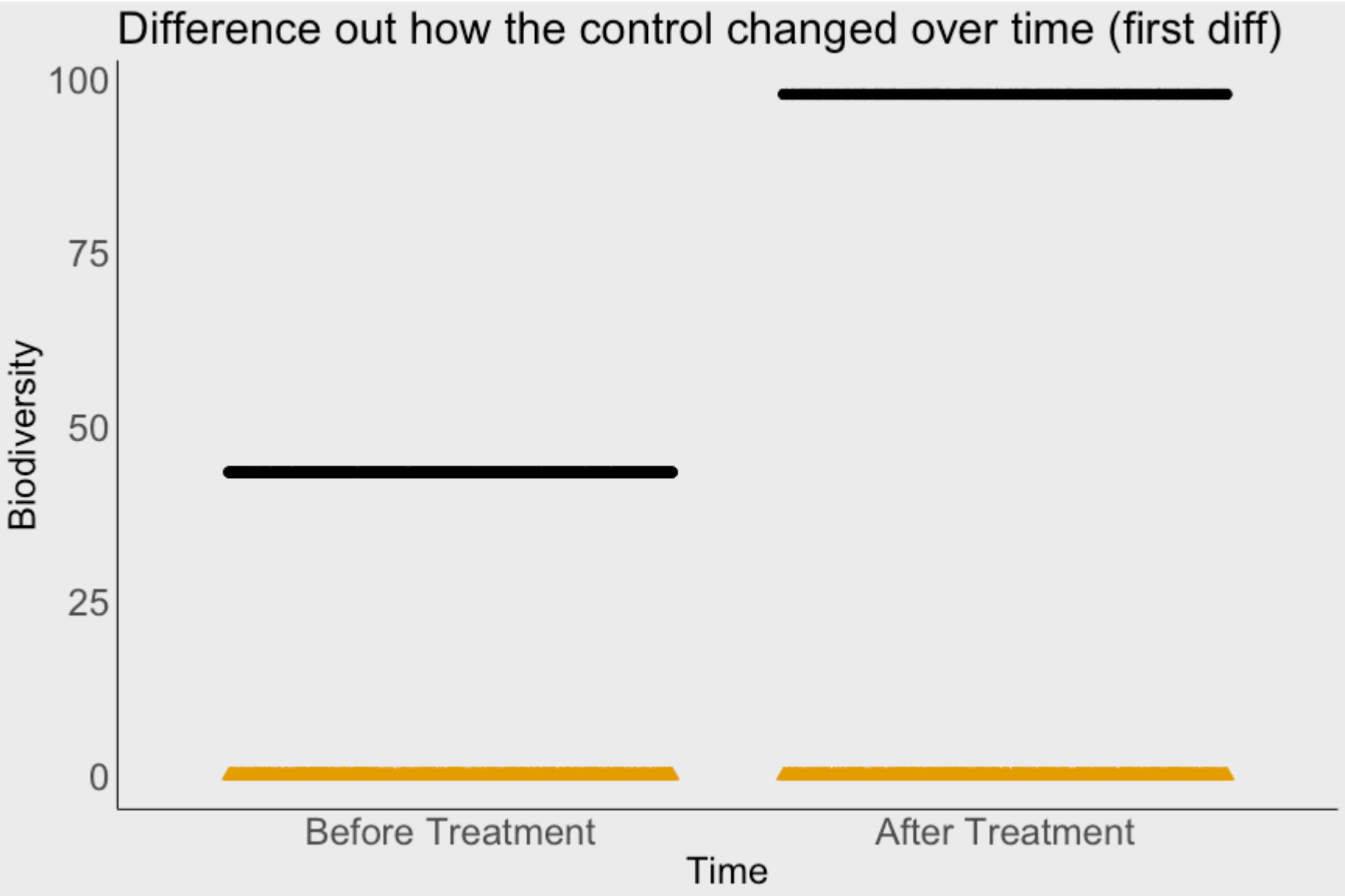
DD step 2


Take the mean of each state-period
500
400
300
200
100
Biodiversity
Before Treatment
After Treatment
Time

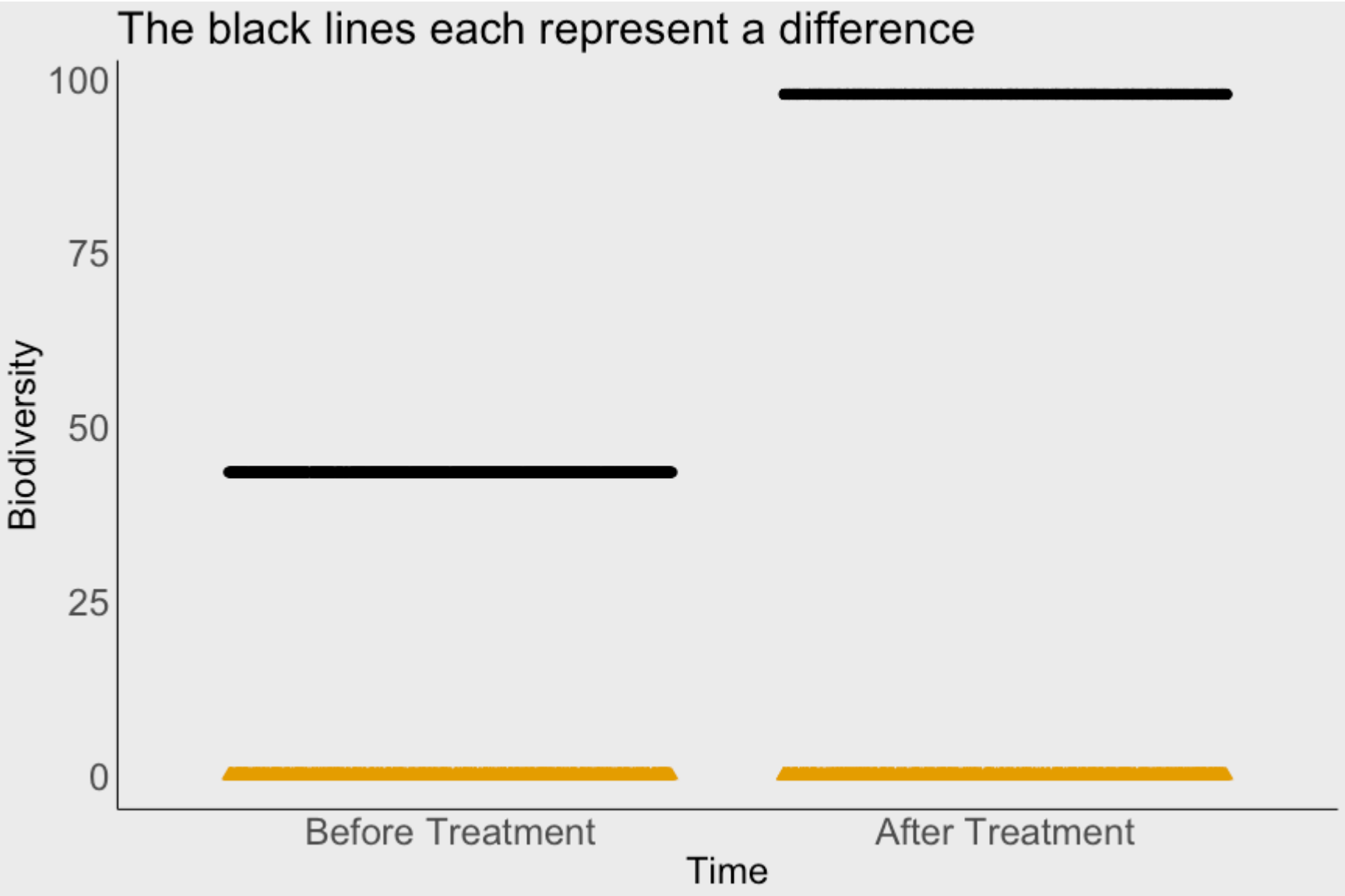
DD step 3


Here are the differences in means for NY and PA
500
400
300
200
100
Biodiversity
Before Treatment
After Treatment
Time

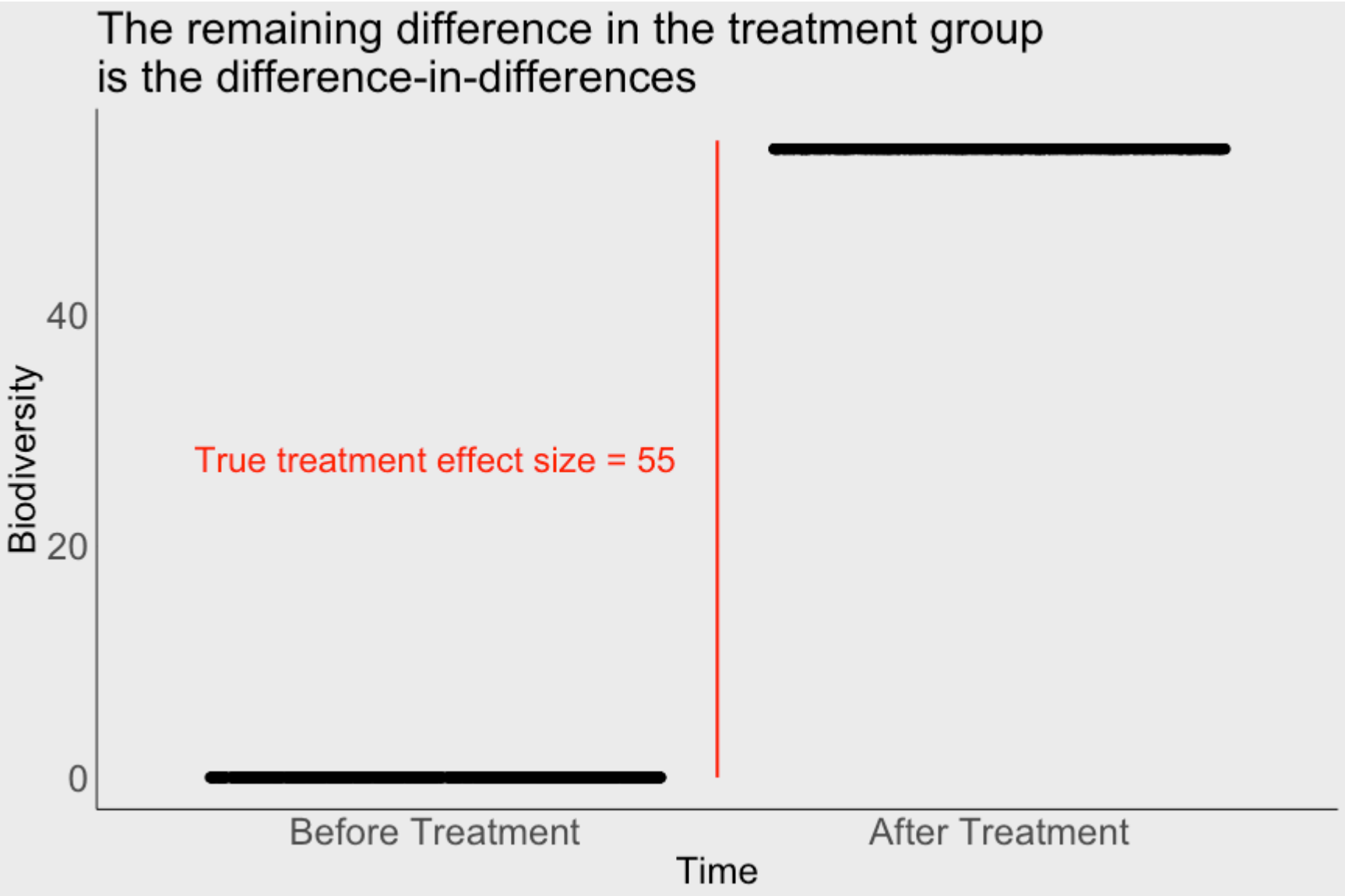
DD step 4


Difference out how the control changed over time (first diff)
100
75
50
25
0
Biodiversity
Before Treatment
After Treatment
Time

DD step 5


The black lines each represent a difference
100
75
50
25
0
Biodiversity
Before Treatment
After Treatment
Time

DD step 6


The remaining difference in the treatment group
is the difference-in-differences
Biodiversity
40
20
0
True treatment effect size = 55
Before Treatment
After Treatment
Time

The effect of leaded gasoline on elderly mortality: Evidence from regulatory exemptions

What is the paper about?

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Why is this important? We know lead is bad

1. There is little causal evidence for any effects of lead
2. Almost zero *causal* evidence for effects of lead on adults in any way
3. Having an accurate measure of effects/costs is vital for policymaking

Environmental research directly affects policy!

PUBLISHED DOCUMENT

AGENCY:

Environmental Protection Agency (EPA).

ACTION:

Notice; call for information.

SUMMARY:

The Environmental Protection Agency (EPA) is preparing an Integrated Science Assessment (ISA) as part of the review of the primary and secondary National Ambient Air Quality Standards (NAAQS) for Lead (Pb). The ISA will be completed by EPA's Office of Research and Development's Center for Public Health and Environmental Assessment (CPHEA). When final, the ISA is intended to update the previous Pb ISA (EPA/600/R-10/075F), published on June 26, 2013. Interested parties are invited to assist EPA in developing and refining the scientific information base for the review of the Pb NAAQS by submitting research studies and data that have been published, accepted for publication, or presented at a public scientific meeting since January 1, 2011.

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Environmental Protection Agency

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Docket ID No. EPA-HQ-OAR-2020-0312
FRL-10011-92-ORD

Document Number:

How does the paper do it?

HR estimates the causal effect of lead by exploiting a **quasi-experiment**

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A quasi-experiment is a real world occurrence that approximates an actual randomized control trial; quasi-experiments are also called natural experiments

How does the paper do it?

Randomly assigning lead exposure to different groups is unethical, but we can learn from situations where real world exposure was *as good as random*

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The quasi-experiment HR exploits is the sudden removal of lead from racing gasoline in 2007

Places that happened to have racetracks in 2007 had a significant decrease in lead emissions relative to places without racetracks

How does the paper do it?

Here's the 2x2 DD table

	Before	After
Treated	**Areas near NASCAR racetracks before 2007**	**Areas near NASCAR racetracks after 2007**
Control	Areas far from NASCAR racetracks before 2007	Areas far from NASCAR racetracks after 2007

We are comparing areas **close** vs far from racetracks, before vs after deleading in 2007

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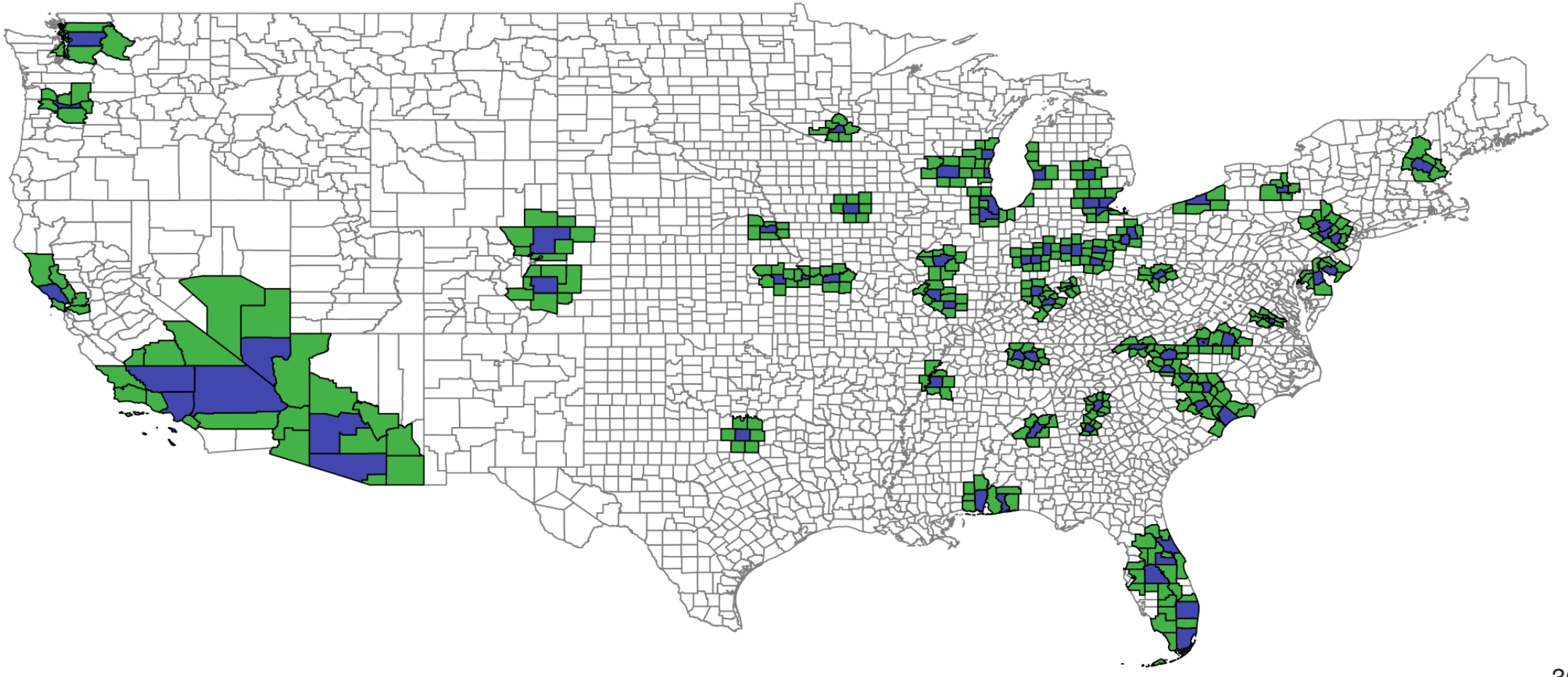
track/treated counties, border/weak treated counties, white control counties



Event studies

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1. We can look at dynamic effects
2. It provides supporting evidence for the parallel trends assumption

Event studies: dynamic effects

With an event study we can see how the effect of a policy changes *over time*

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With an event study we can see how the effect of a policy changes *over time*

We will be able to estimate the effect in each post-period relative to some period

Event studies: dynamic effects

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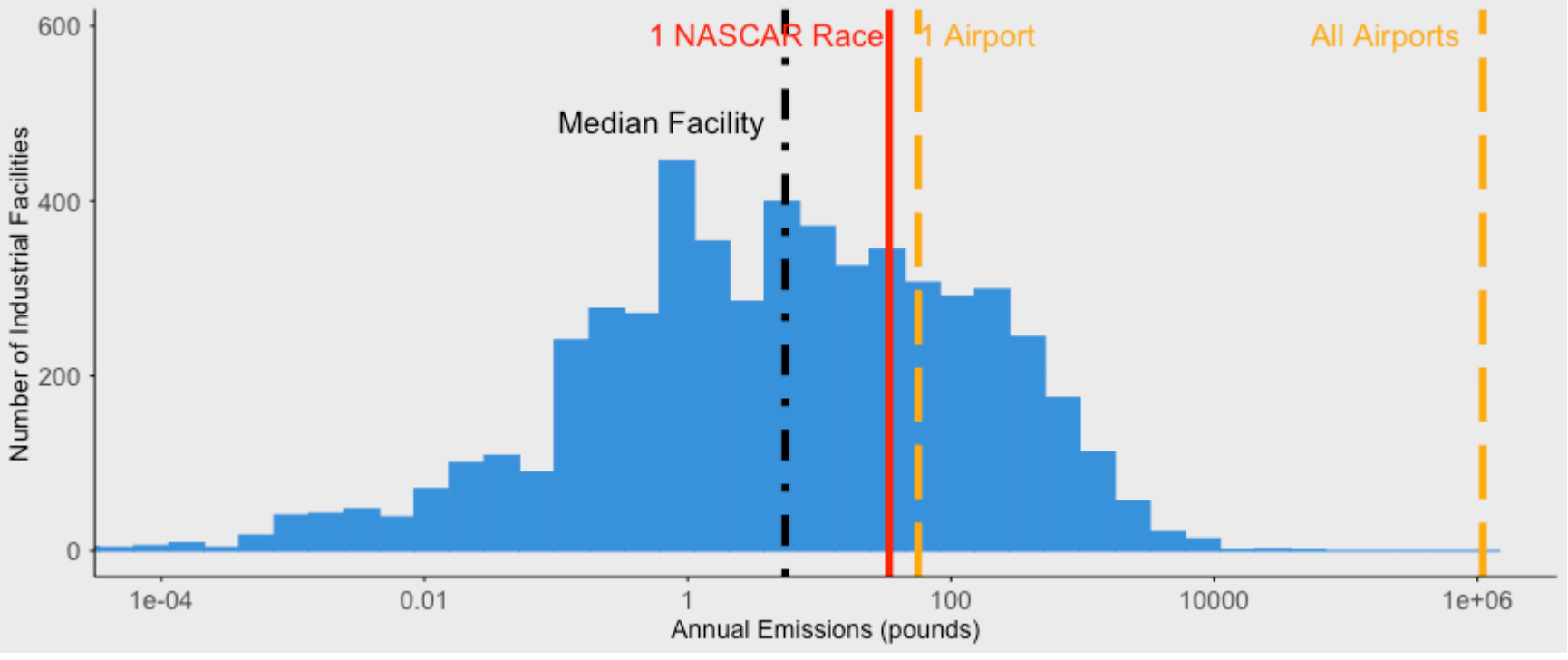
We never observe what would have **actually** happened though, parallel pre-trends is just supporting evidence that the parallel trends assumption holds

HR (2021): to the data

Now lets use the actual HR (2021) data to get some practice: notebook here

NASCAR emitted A LOT of lead

In **3 hours**, one NASCAR race emits more lead than a majority of industrial facilities do in an **entire year**



HR (2021): takeaways

What can we take away from HR (2021)

1. Lead has a causal effect on elderly mortality
   - It appears to be through cardiovascular channels, perhaps *deaths of despair* as well (Case and Deaton, 2015)
2. The mortality costs are large
   - The external mortality costs imposed by NASCAR are larger than the value of all NASCAR teams combined
   - The social cost of lead is **at least $1,000 per gram**
   - This is very, very, very big